Forecasting Swiss Natural Hazards in Graubünden

Final Project Report

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Abstract

Accurate prediction of heavy precipitation and snowfall events is critical for infrastructure planning, risk management, and civil protection in Alpine regions. This study develops a multi-label classification system that predicts the occurrence of heavy rain and heavy snow hazards approx. 19–24 hours in advance using hourly meteorological reanalysis data from the Open-Meteo API for the canton of Graubünden, Switzerland. The dataset spans 56 years (1970–2026) at hourly resolution, comprising approx. 494,500 observations across 27 raw meteorological variables.

The prediction task is framed as a multi-label binary classification problem with severe class imbalance, where heavy rain events occur in only 1.6% of observations and heavy snow in 3.1%. A Logistic Regression baseline is compared against a single-layer LSTM model whose architecture was selected through a 20-trial random hyperparameter search. Both models are trained on data through 2020, with thresholds calibrated on a two-year validation period (2021–2022) covering all seasons, and evaluated on a fully held-out test set spanning 2023 to mid-2026.

Contents

1 Project Objectives

This project addresses the question of whether a multi-label LSTM classifier, trained on historical hourly reanalysis data, can provide operationally useful predictions of heavy rain and heavy snow events in Graubünden that outperform a simple logistic regression baseline at a lead time of approx. 19–24 hours. The prediction task is defined as binary classification for each hazard type: given the meteorological state at time t and the preceding 23 hours, the model predicts whether a heavy rain or heavy snow event will occur in a six-hour window centred around $t + 24$ hours.

The study pursues four primary objectives. First, it constructs a dataset from Open-Meteo reanalysis data with appropriate feature engineering for temporal modelling, including rolling statistics that capture meteorological trends. Second, it conducts exploratory data analysis to characterise the class imbalance, temporal structure, and feature relevance of the hazard prediction task. Third, it implements and evaluates both a Logistic Regression baseline and an LSTM model (Yadav and Thakkar, 2024) using leakage-free time-series cross-validation with hyperparameters selected through random search. Fourth, it quantifies the uplift of the tuned LSTM over the baseline and critically assesses the uncertainty and limitations of both approaches.

The expected outputs of the project are:

1. **Empirical model comparison** – The project delivers quantitative evidence on whether temporal sequence modelling with an LSTM provides measurable predictive uplift over a Logistic Regression baseline. The comparison clarifies for which hazard type—rain or snow—temporal information is most useful and whether the additional model complexity is justified by improved discrimination.

2. **Realistic performance assessment** – Predicting natural hazards at a lead time of approx. 24 hours is inherently challenging, particularly in Swiss Alpine regions where mesoscale weather systems develop rapidly. Both models are expected to achieve strong discrimination (high AUC) but only moderate operational F1 scores, reflecting the fundamental precision–recall trade-off in rare event prediction where high recall comes at the cost of elevated false alarm rates.

2 Data

The dataset is sourced from the Open-Meteo Historical Weather API, which provides hourly meteorological reanalysis data derived from the ERA5 reanalysis product and other numerical weather model outputs. The data is freely available under the Attribution 4.0 International (CC BY 4.0) licence (Zippenfenig, 2023). A single CSV file was downloaded for the Graubünden region, defined by the bounding box 46.46°N–46.72°N, 9.28°E–9.46°E, at approx. 1,500 metres elevation, covering the period from 1 January 1970 to 1 June 2026. The file contains 494,544 hourly observations.

Data Quality Assessment

The data contains sporadic missing values, primarily in soil variables and snow depth during the earliest decades. After forward-fill imputation per location respecting temporal order, followed by back-fill for leading missing values, zero missing values remain in the numeric columns. This imputation strategy is physically justified: meteorological variables typically change slowly relative to the one-hour sampling interval, making the last known value a reasonable estimate for short gaps. For longer gaps in the early record before 1979, prior to ERA5 coverage, back-fill from the first available value provides a conservative fallback.

The hourly series is contiguous with no missing timestamps. The period from 1 January 1970 to 1 June 2026 spans 20,606 calendar days (including 14 leap days), yielding an expected 20,606 $\times$ 24 = 494,544 hourly records. As a caveat, the data is model-derived rather than directly observed, which introduces smoothing bias, particularly for extreme events. Precipitation extremes may be underestimated compared to gauge observations. This is a known limitation of ERA5-based products in complex Alpine terrain (Hersbach et al., 2023).

Feature Name	**Unit**	**Description**
Air Temperature (2 m)	°C	Ambient air temperature measured at 2 metres above ground.
Apparent Temperature	°C	Perceived or feels-like temperature accounting for environmental conditions.
Dew Point Temperature	°C	Temperature at which air becomes saturated and condensation begins.
Relative Humidity (2 m)	%	Relative amount of moisture in the air at 2 metres.
Vapour Pressure Deficit	kPa	Difference between saturated and actual vapour pressure, indicating atmospheric drying potential.
Total Precipitation	mm	Total precipitation amount, including rain and snow.
Rain	mm	Liquid precipitation amount.
Snowfall	cm	Amount of snowfall accumulated.
Snow Depth	m	Depth of snow cover on the ground.
Weather Code (WMO)	–	Encoded representation of the current weather condition according to WMO standards.
Sea-Level Pressure	hPa	Atmospheric pressure reduced to mean sea level.
Solar Radiation	W/m^2	Incoming shortwave solar radiation at the surface.
Sunshine Duration	s	Duration of direct sunshine during the observation period.
Cloud Cover	%	Fraction of the sky covered by clouds.
Wind Speed (10 m)	km/h	Wind speed measured at 10 metres above ground.
Wind Gusts	km/h	Maximum wind gust speed recorded during the observation period.
Wind Direction	°	Direction from which the wind originates.
Month	–	Calendar month of the observation (1–12).
Day of Year	–	Sequential day within the year (1–366).
Hour	–	Hour of the day (0–23).

Table 1: Meteorological and temporal features used in the dataset.

The prediction targets are binary indicators of heavy weather events. The rain hazard target equals one if any hour in a six-hour window following a 24-hour gap has a WMO weather code indicating heavy or violent rain, specifically codes 63, 65, 66, 67, and 82. The snow hazard target equals one if any hour in that window has a code indicating heavy snowfall, specifically codes 75 and 86.[1] Implementation uses a backward-looking rolling maximum after shifting, yielding an effective prediction window of approx. $[t + 19, t + 24]$ hours. The gap ensures the model cannot exploit persistence in the current weather state to trivially predict the near future. The six-hour aggregation window increases the effective positive rate while remaining operationally relevant.

The resulting class distribution shows 7,935 positive cases for rain out of 486,579 negative cases, representing a 1.60% positive rate. For snow, there are 15,452 positive cases out of 479,062 negative cases, representing a 3.12% positive rate. The final 30 rows of each location's time series are trimmed because the future window cannot be fully constructed.

Feature Engineering

Beyond the raw meteorological variables, the modelling pipeline applies three categories of feature transformations. First, cyclical encoding replaces month, day of year, and hour with their sine and cosine representations, yielding six features and preventing the model from treating December-to-January transitions as discontinuities. Second, rolling statistics are computed for five key variables (sea-level pressure, relative humidity, cloud cover, precipitation, and wind gusts): 6-hour rolling mean, 12-hour rolling mean, and 6-hour difference (tendency), adding 15 trend features. These rolling features are strictly backward-looking and do not introduce data leakage. Third, standardisation fits a StandardScaler on training data only and applies it across all timesteps

[1] WMO code table 4677 present-weather codes: 63 = continuous moderate rain; 65 = continuous heavy rain; 66 = slight freezing rain; 67 = moderate or heavy freezing rain; 82 = violent rain shower(s); 75 = continuous heavy snowfall; 86 = moderate or heavy snow shower(s).

in the flattened sequence matrix. The final feature set for the LSTM comprises 32 dimensions: 11 meteorological variables, 6 cyclical time features, and 15 rolling statistics.

3 Exploratory Data Analysis

The primary meteorological features over the full 56-year period reveal the characteristics of Graubünden's high-altitude continental climate. Air temperature at 2 metres ranges from -32.5°C to 27.7°C with a mean of 1.16°C and standard deviation of 8.47°C. Apparent temperature shows similar patterns, ranging from -37.9°C to 27.4°C with a mean of -1.71°C. Relative humidity averages 79.65% with substantial variability. Precipitation averages 0.17 mm per hour with maximum values reaching 10.5 mm, while snowfall averages 0.06 cm per hour with peaks at 5.18 cm. Sea-level pressure averages 1018.5 hPa with low variability. Cloud cover averages 69.7% with a median of 88%, indicating a predominantly overcast distribution. Wind speed averages 5.26 km/h while gusts reach maxima of 127.1 km/h, indicating föhn-type events characteristic of inner-Alpine locations.

Event Duration

Analysis of consecutive hours with hazard conditions reveals distinct patterns for each hazard type.

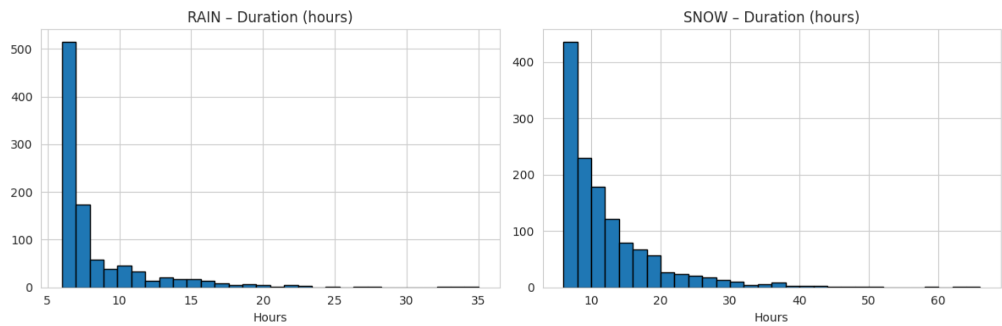

Figure 1: Count of events by duration hour.

For rain, 986 distinct events occur over the 56-year period, averaging 18 per year. These events have a median duration of 6 hours, a mean of 8.0 hours, a maximum of 35 hours, and a 90th percentile of 13 hours. For snow, 1,312 distinct events occur, averaging 23 per year. Snow events are longer with a median duration of 9 hours, a mean of 11.8 hours, a maximum of 66 hours, and a 90th percentile of 20 hours. Snow events are longer and more frequent, consistent with synoptic-scale frontal forcing, while rain includes shorter convective episodes. The overlap between rain and snow hazards is minimal at only 278 hours or 0.056% of all observations, confirming that the targets are largely independent. The 24-hour lookback window fully captures the temporal context of most events.

Class Imbalance

The target variables exhibit severe imbalance. Monthly analysis reveals strong seasonality: rain concentrates in summer months, with July showing a 5.36% positive rate, August 5.09%, and June 3.93%, while winter months show essentially zero rain hazard events. Snow hazard events peak in November at 6.34% and April at 5.99%, with minimal summer occurrence of only 0.47% in July.

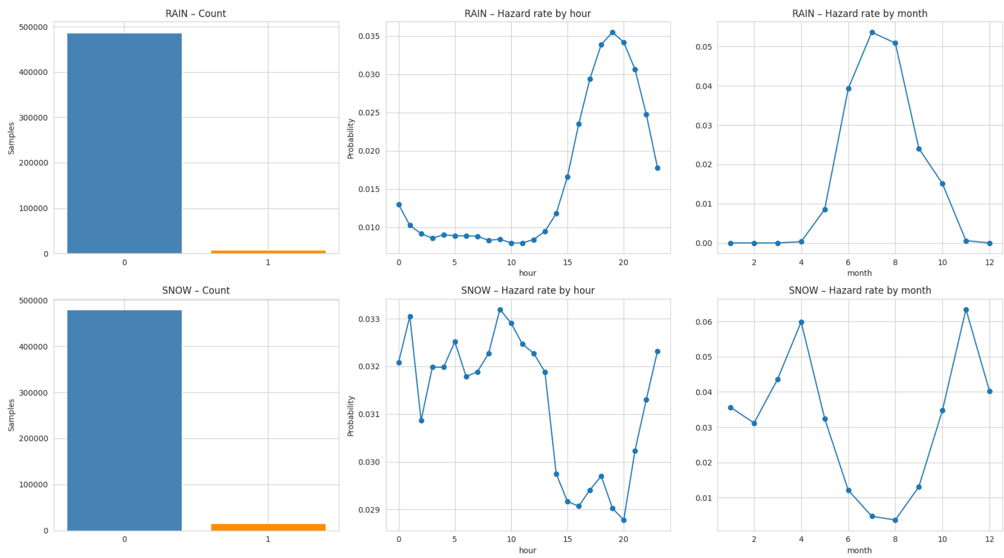

Figure 2: Distribution of hazardous events.

Hourly analysis shows weak diurnal patterns. Rain peaks at hour 19 with a 3.55% rate, consistent with afternoon convective activity, versus a minimum at hour 10 of only 0.80%. Snow varies minimally between 2.88% and 3.32% across hours, consistent with synoptic-scale rather than convective forcing. This seasonality motivated the choice of a two-year validation set (2021–2022) covering all seasons, ensuring that threshold calibration accounts for both summer rain and winter snow patterns.

Feature–Target Correlations

Pearson correlations between instantaneous features and the prediction targets reveal the relationships available to the models. For rain hazard, the strongest positive correlations are with apparent temperature ($r = 0.160$), air temperature at 2 metres ($r = 0.156$), and current rain ($r = 0.112$). Current precipitation shows a weaker correlation of 0.066, while wind direction shows 0.061. Sea-level pressure shows a weak negative correlation of -0.030, and snowfall shows -0.033.

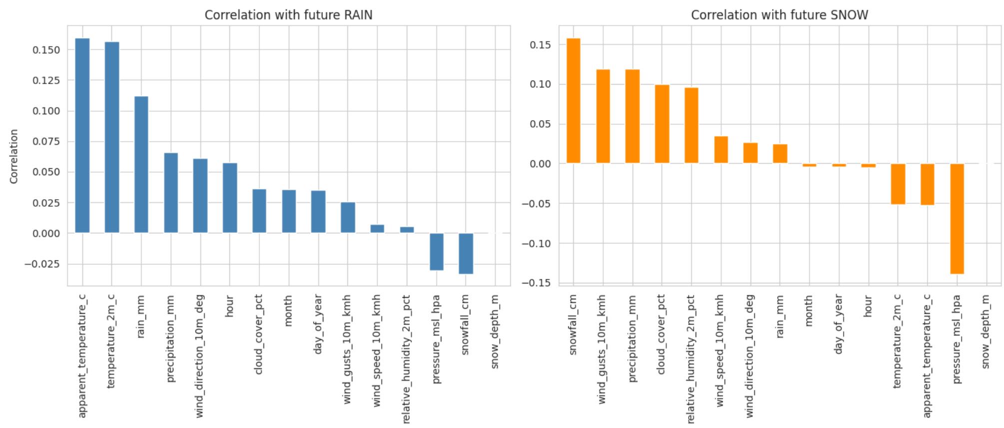

Figure 3: Correlation of features with target variable.

For snow hazard, the pattern differs substantially. Current snowfall shows the strongest correlation ($r = 0.159$), followed by wind gusts ($r = 0.119$), precipitation ($r = 0.119$), cloud cover ($r = 0.100$), and relative humidity ($r = 0.096$). Apparent temperature shows a weak negative correlation of -0.053, while sea-level pressure shows a stronger negative relationship ($r = -0.139$), indicating that low-pressure systems precede snow events.

All correlations are modest with absolute values below 0.16, confirming that no single feature provides strong linear prediction at the 24-hour lead time.

Temporal Autocorrelation

Partial autocorrelation function analysis at 48 lags shows significant autocorrelation up to 12–18 hours for both targets. Cross-correlation analysis reveals that certain features lead the targets by characteristic intervals: pressure and humidity lead snow by 12–24 hours, while temperature and precipitation lead rain by 6–12 hours. These patterns validate the choice of a 24-hour lookback window and provide direct motivation for the rolling statistics (6-hour and 12-hour means, 6-hour tendency) included as engineered features.

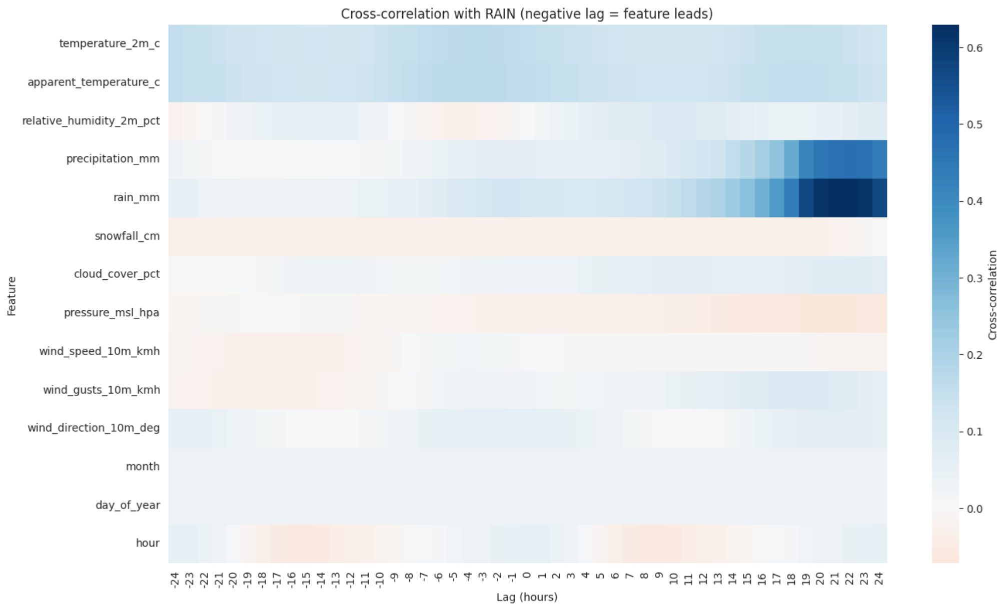

Figure 4: Cross-correlation for rain (negative lag = feature leads).

Based on the exploratory analysis, three hypotheses guide the modelling effort. First, sequence-based models such as LSTMs should outperform instantaneous models due to the observed temporal autocorrelation and feature lead-time patterns. Second, snow prediction should achieve higher AUC than rain prediction due to stronger feature correlations and longer event durations. Third, class-weighted loss functions and threshold optimisation should prove essential for achieving operationally useful recall at acceptable precision.

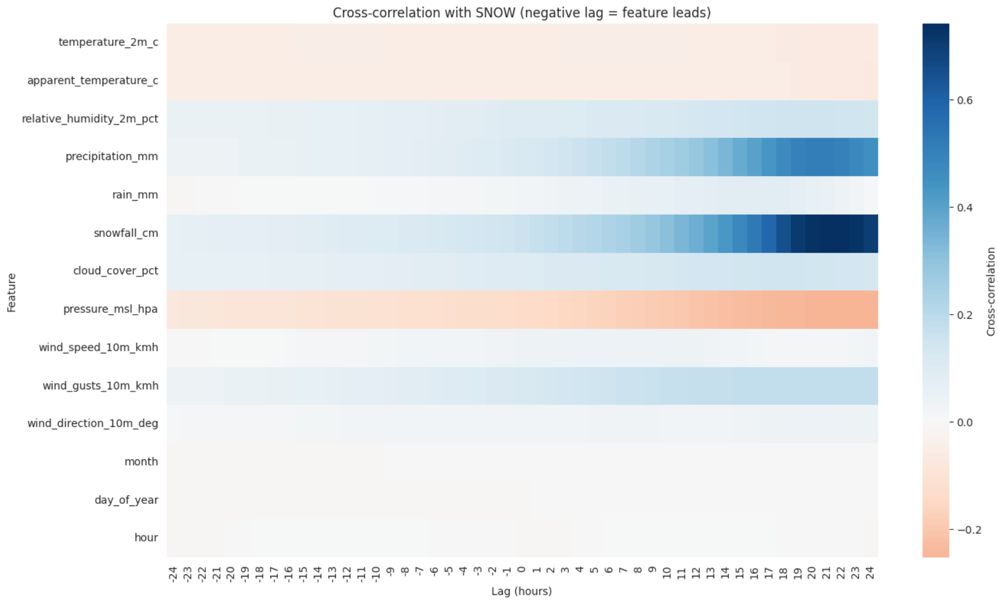

Figure 5: Cross-correlation for snow (negative lag = feature leads).

4 Modelling Approach

The modelling task is multi-label binary classification. The input consists of meteorological features at time t and the preceding 23 hours. The output consists of the probabilities of a rain hazard and a snow hazard occurring in a six-hour window at approx. 24 hours lead time. The two targets are modelled jointly through a shared LSTM encoder with independent output heads and per-class threshold optimisation.

Baseline Model

The baseline serves as a reference point to quantify the value of temporal modelling. The model takes as input only the last timestep of each 24-hour window, yielding 32 features (11 meteorological, 6 cyclical, 15 rolling statistics). Two independent binary classifiers are trained, one for each hazard type. Both use balanced class weights, which effectively oversample the minority class during optimisation. The solver is L-BFGS with a maximum of 1,000 iterations. Decision thresholds are optimised per-class on the validation set by maximising F1 score.

Logistic Regression provides an appropriate baseline because it is interpretable, computationally efficient, and provides a strong linear reference point. Although it receives the rolling features (which encode recent trends), it processes only a single time point, directly testing whether the full 24-hour temporal sequence adds value beyond what is captured in engineered trend features (Foresti et al., 2019).

LSTM Model

The LSTM architecture processes the full 24-hour sequence. Input sequences have shape (batch_size, 24 timesteps, 32 features). These pass through a single-layer LSTM with 64 hidden units. The last hidden state is extracted and passed through a classification head consisting of dropout at 0.40, a linear layer reducing from 64 to 32 units, ReLU activation, a second dropout at 0.40, and a final linear layer producing two output logits.

The architecture was selected through a 20-trial random hyperparameter search (see Appendix). Single-layer architectures dominate the top ranks, suggesting that the data constrains model capacity more than architectural depth. Higher dropout (0.30–0.40) consistently outperforms lower regularisation, indicating overfitting risk on this imbalanced dataset. The best configuration achieves a validation macro F1 of 0.4945 and macro AUC of 0.9149.

The loss function is binary cross-entropy with logits, using positive class weights equal to the ratio of negative to positive training samples— approx. 63 for rain and 30 for snow. Training uses the Adam optimiser with learning rate 0.0005 and batch size 32. Early stopping monitors validation loss with patience of 5 epochs. The model produces raw logits, and sigmoid activation yields per-class probabilities. Rather than using the default threshold of 0.5, which is inappropriate for imbalanced problems, the study sweeps thresholds from 0.05 to 0.95 in steps of 0.01 on the validation set and selects the threshold maximising the F1 score for each class independently. This approach balances high recall for hazard detection against acceptable precision to limit alarm fatigue.

Training Strategy

Sequence creation uses stride 6 for training data (yielding approx. 74,500 sequences from 447,000 hourly rows), which reduces redundancy between overlapping windows and prevents the immediate overfitting observed with stride 1 (460,000+ highly correlated sequences). Validation and test data use stride 1 for maximum resolution during threshold calibration and evaluation.

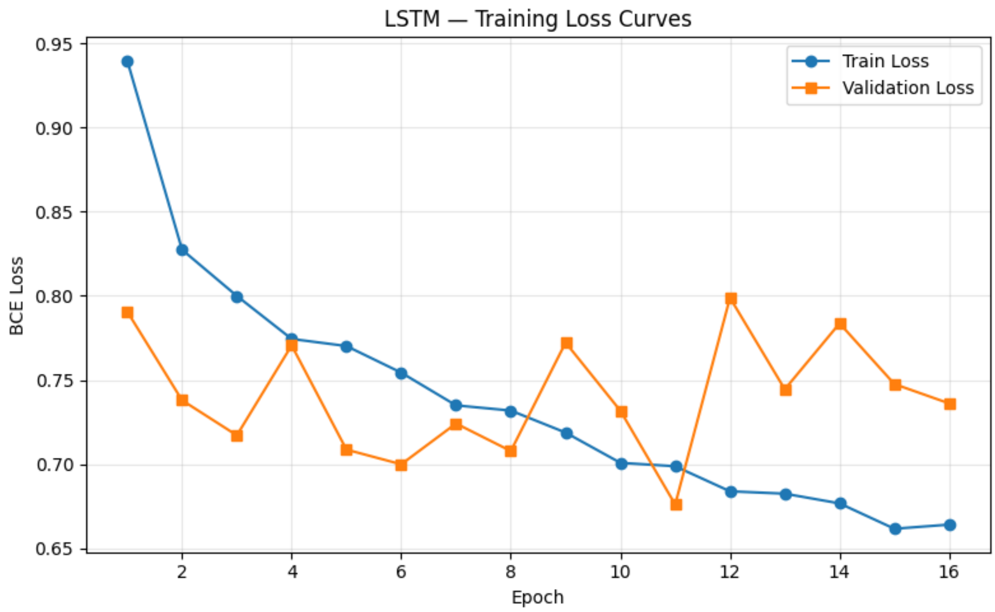

Figure 6: Loss curves for training and validation sets.

Validation Approach

Time-series data requires special care to avoid look-ahead bias. The study employs sliding-window cross-validation with four folds over the training period (1970–2020). Each fold maintains a minimum two-year training window before the first validation block. StandardScaler is fit on training data only within each fold, and thresholds are optimised per fold.

For final evaluation, the data is split as follows:

- **Training set**: All data through December 2020 (approx. 51 years, 447,000 rows)
- **Validation set**: January 2021 through December 2022 (2 full years, 17,500 rows)
- **Test set**: January 2023 through June 2026 (approx. 3.4 years, 29,900 rows)

The two-year validation period was chosen to cover all seasons twice, ensuring that threshold calibration accounts for both summer rain patterns and winter snow patterns. In line with Lang et al., 2023 with an earlier configuration using only October–December 2022 (winter-concentrated) achieved high validation F1 but poor test generalisation due to seasonal distribution mismatch.

Evaluation Metrics

Given the severe class imbalance, accuracy is not a meaningful metric—a naive model predicting negative for all samples would achieve 98.4% accuracy for rain. The primary metric is AUC-ROC, which measures threshold-agnostic discrimination ability. F1 score provides a single metric combining precision and recall at the optimised threshold. Precision and recall are reported separately for operational interpretation. Confusion matrices show absolute error counts. Macro-averaged AUC provides a single summary across both hazard types.

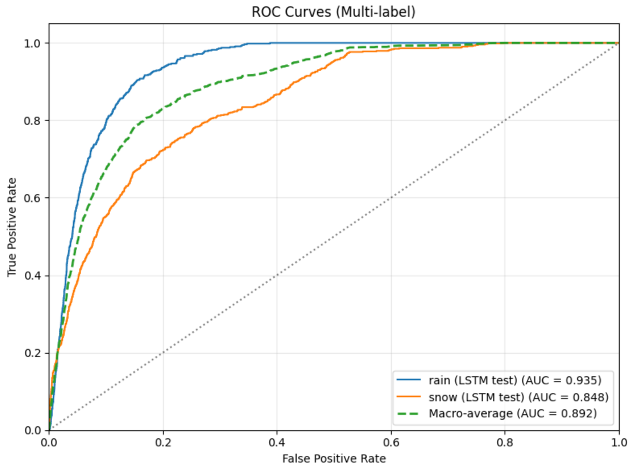

Figure 7: ROC-Curve for tuned LSTM model.

5 Results

This section presents the predictive performance of the Logistic Regression baseline and the tuned LSTM model on the held-out test period from January 2023 to June 2026, comprising 29,898 evaluation sequences. Performance is reported separately for heavy rain and heavy snow, using AUC-ROC as the primary threshold-independent metric and F1 at the validation-optimised threshold as the operational metric.

Cross-Validation

Four-fold sliding-window cross-validation over the training period (1970–2020) yields a mean macro AUC of 0.875 ± 0.028. Performance improves monotonically with training set size: Fold 1 (2 years training) achieves 0.827, while Fold 4 (39 years) achieves 0.895. This confirms that the LSTM benefits from longer historical records and that the architecture seem to generalises stronger across different temporal regimes.

Logistic Regression Baseline

For the rain hazard, the baseline achieves an AUC of 0.9182 on the test set. At the validation-optimised threshold of 0.75, it obtains precision of 0.11, recall of 0.78, and F1 of 0.1926. The confusion matrix shows 442 true positives and 122 missed events out of 564 rain positives in the test period, with 3,585 false alarms.

For the snow hazard, the baseline achieves an AUC of 0.8101. At a threshold of 0.73, it obtains precision of 0.10, recall of 0.42, and F1 of 0.1587. The model correctly identifies 304 of 729 snow events while generating 2,798 false alarms. Across both hazard types, the baseline obtains a macro-averaged AUC of 0.8642.

Tuned LSTM

The LSTM was trained for 16 epochs before early stopping (best model at epoch 11). Final training used positive class weights of approx. 63 for rain and 30 for snow. For the rain hazard, the LSTM achieves an AUC of 0.9354 on the test set. At the validation-optimised threshold of 0.89, it obtains precision of 0.19, recall of 0.56, and F1 of 0.2829. The confusion matrix shows 314 true positives and 250 missed events, with only 1,342 false alarms—less than half the baseline's false alarm rate.

For the snow hazard, the LSTM achieves an AUC of 0.8480. At a threshold of 0.71, it obtains precision of 0.12, recall of 0.56, and F1 of 0.1973. The model correctly identifies 405 of 729 snow events with 2,971 false alarms. Across both hazard types, the LSTM obtains a macro-averaged AUC of 0.8917.

Model Comparison

The tuned LSTM outperforms the Logistic Regression baseline on all metrics. The LSTM's improvement is most pronounced for rain, where F1 increases by 47% relative to the baseline (0.283 vs. 0.193). This improvement is driven primarily by a substantial reduction in false alarms (1,342 vs. 3,585) at the cost of moderately lower recall (0.56 vs. 0.78). For snow, the improvement is more modest. The LSTM gains 3.8 points in AUC and 3.9 points in F1, suggesting that temporal patterns provide less incremental value for snow prediction, which is already well-characterised by instantaneous features such as pressure, humidity, and temperature.

Hazard	**Model**	**AUC**	**F1**	**Δ vs. LR**
Rain	Logistic Regression	0.9182	0.1926	–
Rain	Tuned LSTM	0.9354	0.2829	+0.017 AUC / +0.090 F1
Snow	Logistic Regression	0.8101	0.1587	–
Snow	Tuned LSTM	0.8480	0.1973	+0.038 AUC / +0.039 F1
Macro	Logistic Regression	0.8642	–	–
Macro	Tuned LSTM	**0.8917**	–	**+0.028 AUC**

Table 2: Test set performance comparison. Thresholds calibrated on 2021–2022 validation period.

Validation metrics closely track test performance: rain val F1 = 0.300 vs. test F1 = 0.283; snow val F1 = 0.216 vs. test F1 = 0.197. This close alignment confirms that the two-year all-season validation strategy produces thresholds seem to generalise to unseen data.

6 Discussion

The results confirm the primary hypothesis: a tuned LSTM exploiting 24-hour temporal sequences provides measurable discriminative improvement over a Logistic Regression baseline that operates on instantaneous features. The macro AUC improvement of 2.75 percentage points (0.892 vs. 0.864) is consistent across both hazard types and across cross-validation folds, suggesting that this uplift reflects genuine learning of temporal precursor patterns rather than overfitting.

The improvement is most substantial for rain prediction (+4.7% relative F1 gain, +1.7 points AUC), which is somewhat surprising given that rain events are shorter and more convective in character. One plausible explanation is that the rolling features—particularly the 6-hour pressure tendency and humidity trends—capture pre-convective atmospheric destabilisation that the raw instantaneous values miss. The LSTM can additionally learn sequences of feature changes (e.g., rapidly falling pressure followed by rising humidity) that signal approaching fronts or thunderstorms.

For snow, the improvement is more modest (+3.8 points AUC, +3.9 points F1). Snow events in Graubünden are predominantly driven by synoptic-scale low-pressure systems, whose signature is already well captured by instantaneous pressure and humidity values. The temporal sequence adds less incremental information when the current atmospheric state already contains most of the predictive signal.

The hypothesis that snow would achieve higher AUC than rain due to stronger feature correlations is contradicted by the results. Rain AUC (0.935) substantially exceeds snow AUC (0.848) for the LSTM. This reversal suggests that the raw Pearson correlations—which measure linear, instantaneous associations—underestimate the predictive power of nonlinear temporal patterns for rain. The LSTM appears to learn precursor sequences for rain that are not visible in static correlations.

Limitations

Several limitations constrain the interpretation and generalisability of these results. The model is trained and evaluated on a single Alpine grid point; spatial generalisation to other locations is untested and would require multi-station data or spatial features.

The selected architecture (hidden=64, 1 layer, dropout=0.40) may not be globally optimal for the current all-season validation, though the cross-validation stability (AUC 0.875 ± 0.028) suggests the architecture is robust. The binary targets provide no information about intensity gradation or spatial extent—a marginal case just exceeding the WMO code threshold is weighted identically to a severe event. Finally, reanalysis-derived labels may misclassify borderline events; no validation against station observations or damage reports was performed.

Despite strong AUC values, F1 scores remain below 0.30. This doesn't seem to be a model failure but a direct consequence of the base rate. With rain positive rate at approx. 1.6% in the test set, even at recall of 0.56, the precision-recall trade-off is unfavourable: for every true detection, the model generates approx. 4 false alarms. An AUC of 0.935 means the model ranks most true events above non-events, but at any fixed operating threshold, the ratio of positives to negatives constrains achievable F1.

Operationally, the relevant question is whether 4–5 false alarms per true detection is acceptable. For high-consequence hazards such as flooding or avalanche risk, this ratio may be tolerable compared to missing events entirely. The high thresholds selected (0.89 for rain, 0.71 for snow) reflect an implicit preference for precision over recall, which may or may not align with operational requirements.

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A Appendix

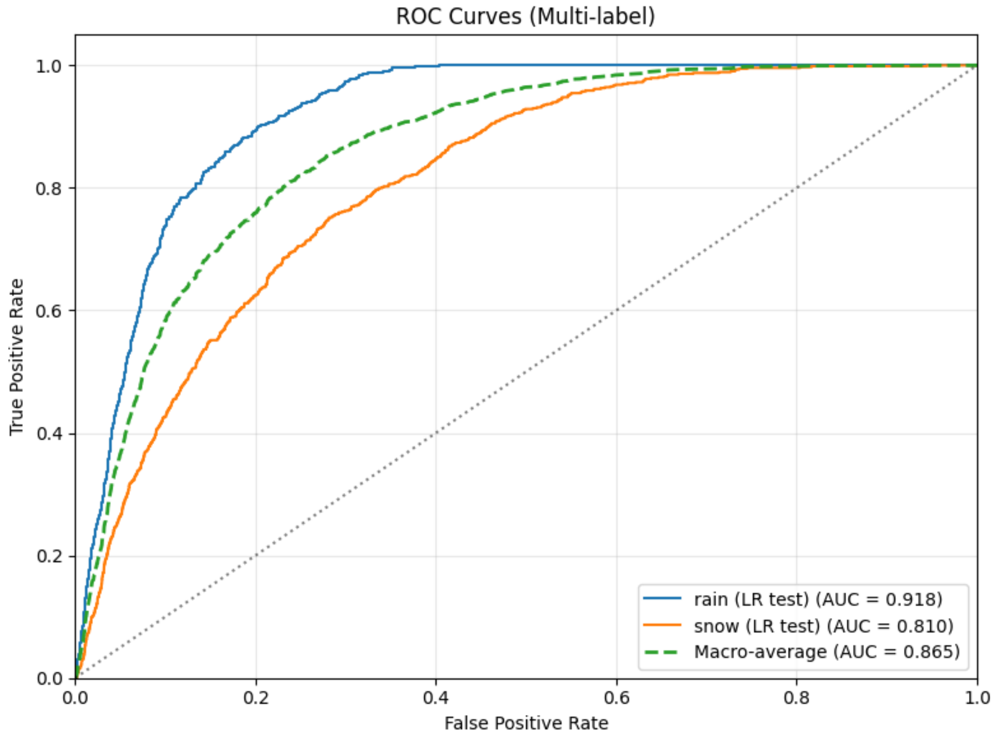

Figure 8: ROC-Curve for logistic regression baseline model.

Table 3: LSTM Hyperparameter Tuning Results (20 Trials, Random Search). Trials ranked by validation macro F1. Training used stride=6, max 25 epochs, patience=5, BCE loss. Feature set: 32 (17 base + 15 rolling).

Rank	**Hidden**	**Layers**	**Dropout**	**LR**	**Batch**	**Macro F1**	**Macro AUC**
1	64	1	0.40	0.0005	32	**0.4945**	0.9149
2	32	1	0.30	0.0010	128	0.4783	0.9250
3	128	1	0.30	0.0020	256	0.4648	0.9107
4	128	3	0.30	0.0010	256	0.4541	0.9146
5	256	2	0.20	0.0010	256	0.4523	0.9074
6	128	1	0.10	0.0020	64	0.4503	**0.9318**
7	256	1	0.10	0.0050	64	0.4388	0.9141
8	64	1	0.20	0.0005	64	0.4377	0.9195
9	256	1	0.30	0.0005	128	0.4269	0.9024
10	256	1	0.40	0.0010	64	0.4261	0.9252
11	256	2	0.40	0.0050	64	0.4258	0.9023
12	256	2	0.20	0.0010	256	0.4231	0.9105
13	32	3	0.30	0.0010	32	0.4026	0.9163
14	64	1	0.30	0.0010	64	0.4017	0.9049
15	256	3	0.20	0.0010	128	0.3990	0.9139
16	64	2	0.20	0.0010	256	0.3939	0.9132
17	256	3	0.20	0.0005	64	0.3874	0.9023
18	256	3	0.40	0.0020	256	0.3816	0.9051
19	64	3	0.40	0.0020	256	0.3739	0.8961
20	128	3	0.30	0.0050	32	0.2974	0.8515

Note: Best configuration (Rank 1) selected for final training at stride=1. Single-layer architectures dominate the top ranks, suggesting the data constrains model capacity more than the architecture. Higher dropout (0.30–0.40) consistently outperforms lower regularisation, indicating overfitting risk on this imbalanced dataset.

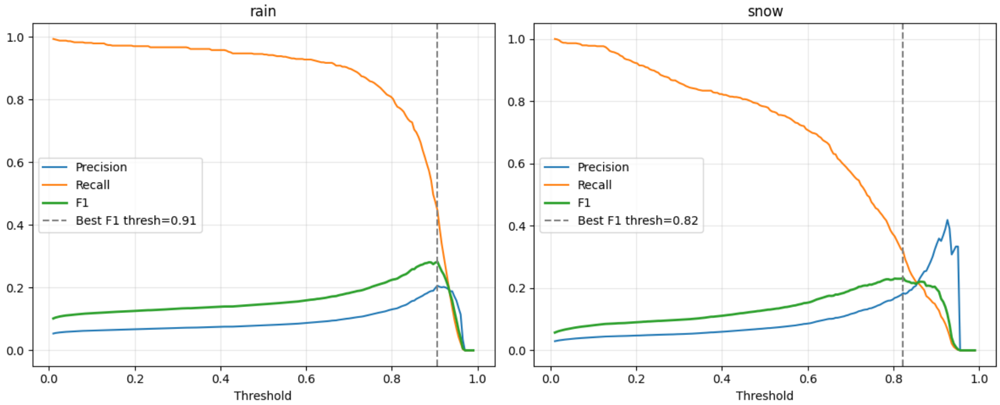

Figure 9: Optimal class threshold for LSTM model.

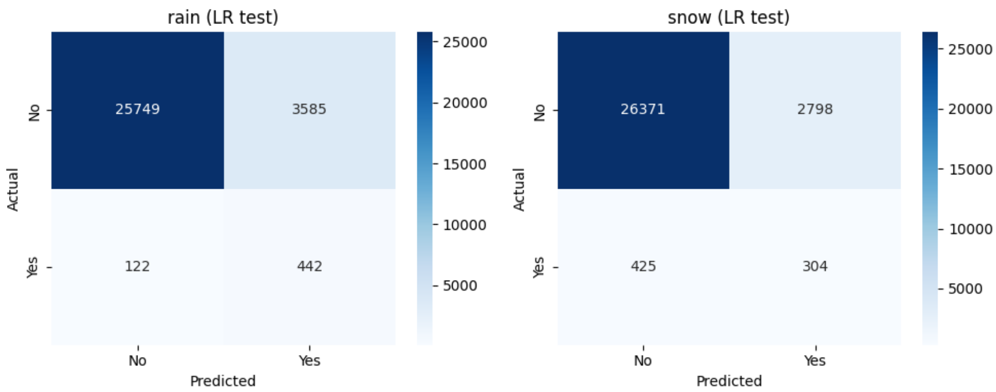

Figure 10: Confusion matrix logistic regression baseline model.

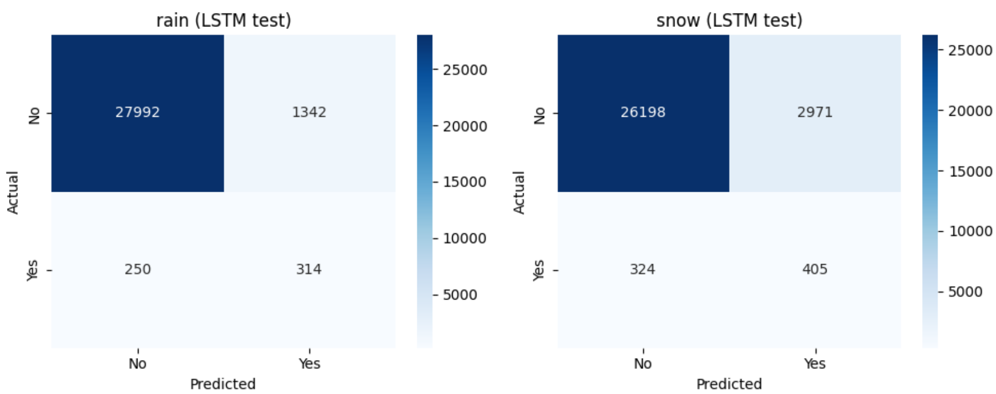

Figure 11: Confusion matrix LSTM model.

Declaration of Authorship

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